Taenacity

BME 1 Group 12

Abigail Marciniak, Brian Pham Ngo, Daniyal Rauf, Imart De Guzman, Isaac Min Bae, Shreya Thota

LFAssociates

LFAssociates

Lateral Flow Assay

- Biotechnology startup
- Operate within medical diagnostics
  - We specialize in Lateral Flow Assay development and production

Our Team

CEO
Abigail Marciniak
5 years in LFA research + industry

Business Analyst
Daniyal Rauf
5 years of industry experience

Marketing Specialist
Brian Pham Ngo
5 years of industry experience

Regulatory Affairs Director
Imart De Guzman
5 years of industry experience

Manufacturing Lead
Isaac Min Bae
5 years in LFA manufacturing

Head of R&D
Shreya Thota
5 years in LFA research

1. Current Disparity

Issue Addressed:

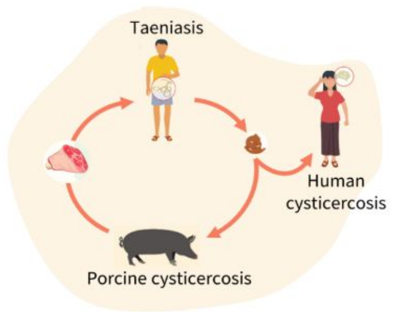


Fig. 3. *T. solium* life cycle
(development of neurocysticercosis)

Taeniasis Solium parasite

Cysticercosis: T. Solium larvae encyst in body tissue-including the brain

-ingestion of eggs from feces of a person w/Taeniasis

Can cause severe neurological symptoms!

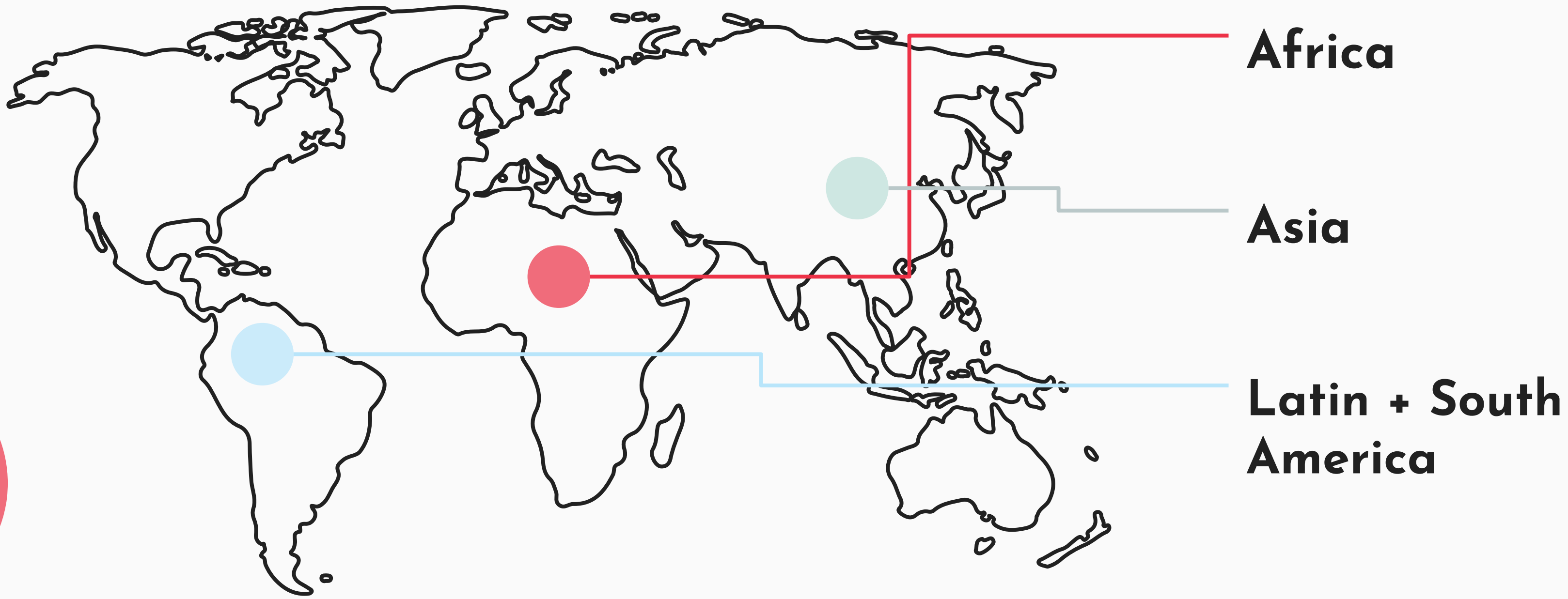
Neurocysticercosis=responsible for 30% of epilepsy cases in endemic areas

Taeniasis: intestinal infestation of mature T. Solium worm

-from consumption of raw or undercooked infected pork

Contributes to spread of cysticercosis

Areas of High Prevalence



Scope: over 2.5 million affected!

2. Market Analysis

Demographic

Target Demographic

Healthcare providers in tropical regions and patients at risk of T. Solium Taeniasis and Cysticercosis.

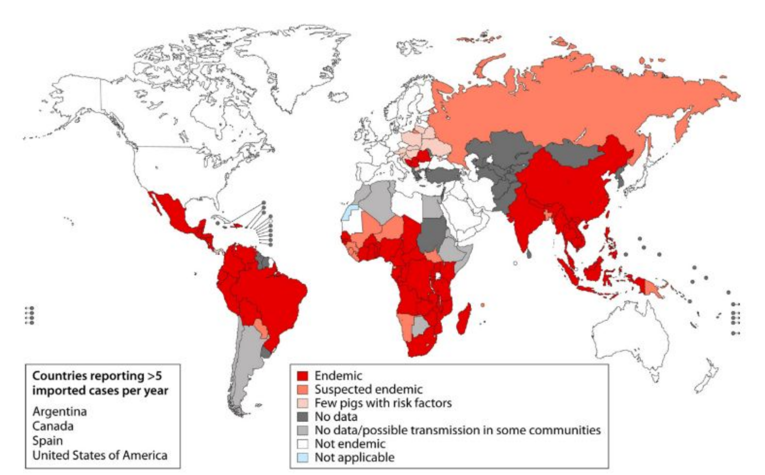


FIG 1 Geographic distribution of *Taenia solium* taeniasis/cysticercosis (WHO, 2015).

Goals

-Improve healthcare accessibility in impoverished tropical regions with a budget-efficient POC device

-Develop accurate and easily accessible POC device for T. Solium Taeniasis and Cysticercosis.

Current Diagnostic Devices

- **<u>No current POC on the market</u>**

- **Current prototype of possible POC:**

  Three-step in-house-produced rapid diagnostic test (RDT) for the simultaneous detection of taeniosis (TST) and cysticercosis (TSCC) antibodies in a lateral-flow format based on two recombinant proteins, rES33 and rT24H, respectively.

  - Sensitivity: 38%, Specificity: 99%

- **Traditional detection strategies:**
  - Stool microscopy
  - Serologic Diagnosis
  - Coproantigen, nucleic acid-based, and serologic testing

All current diagnostic devices/strategies are confined to labs + not freely on the market

3. Device Description

Device Description

Point-of-care capacity:

-No lab testing needed

-Rapid diagnosis

-Benefits patients in remote + underserved regions with limited health infrastructure

Gold nanoshells: Gold nanoshells offer unique optical properties, including high sensitivity to molecular binding events➔***6-20x increase in sensitivity***

-Reduced false negatives

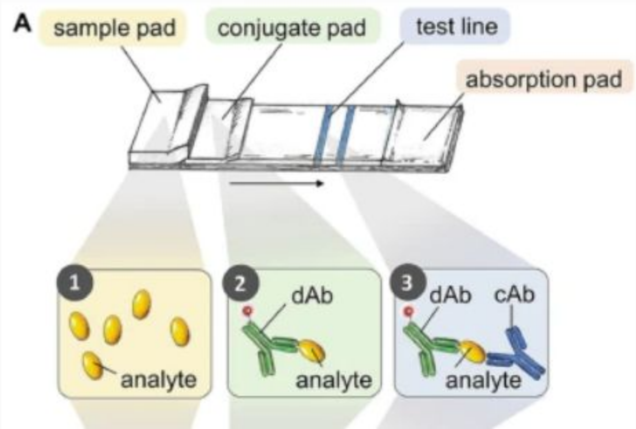
-Prior prototype w/traditional nanoparticles demonstrated only 38% sensitivity

Cost-effectiveness:

-Estimated $20 per test ($3 manufacturing)

-Affordable for healthcare providers/centers in low income areas

Abigail



Key elements of an LFA

Taenacity Features:

-Two embedded test strips, for simultaneous testing of Taeniasis + cysticercosis

-Use of 150 nm gold nanoshells vs. 40 nm nanoparticles (industry standard)

-2 micropipettes for blood sample collection

-Chase buffer w/no need for refrigeration

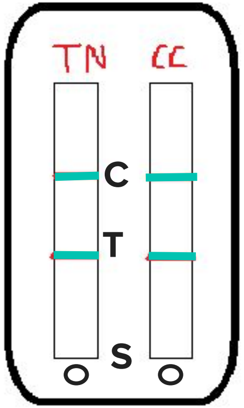
-Easy, primarily visual instructions

Analytes:
rES33 antibodies-Taeniasis
rT24H antibodies-Cysticercosis

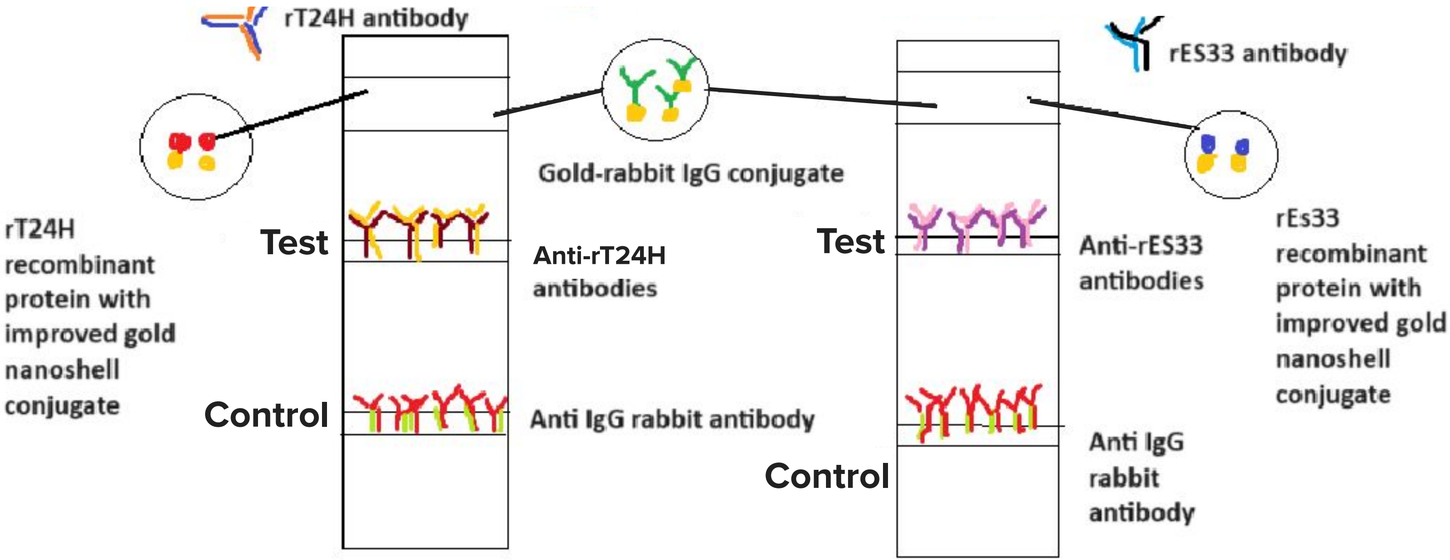
Anti-IgG rabbit antibodies for test control

Abigail

Device Schematic:



Cysticercosis (CC) / Taeniasis (TN)



A B

Chase buffer + micropipette

Abigail

Regulatory Affair

Regulatory Plans

- Adhere to ISO 13485 (international medical device quality control)
- Protocol requires specific physical-chemical characterization, in vitro characterization + in vivo characterization
- Ensure that clinical trials follow the GCP guidelines
- Conduct FTO analysis to assess if the lateral flow test infringes on other current patents
- **FDA Type II Device**
- Type 2 devices have to go through the 510(k) process to get approval from the FDA
- **510(k) Premarket Notification: 510(k) Number: K031942**

Device Testing + IP Protection

Manufacturing/Testing plan:

Manufacturing: Outsource prototyping/product to larger device manufacturer

Prototype testing: Verify there are two distinct lines on the device

Device efficacy (Sensitivity and Specificity):

- Sample size: ~20 end users, ~ 1000 tests with device
- Setting: If budget allows: Eastern Europe, Eastern Africa, or Latin America
- Alternatively: test on patients in US (lower incidence)

IP Protection:

- Trademark IP protection to protect unique "Taenacity" name associated with our product
- US8399261B2 "Lateral flow assay system and methods for its use" is an example of a current patent protecting a LFA

We plan to patent our device

***Critical* Claims:**

1. "A multiplexed LFA device for Taeniasis + Cysticercosis detection

2. Use of rES33 and rT24H antigens, and 150 nm gold nanoshells

Shreya

Marketing Strategy

Medical Device Distributors

Local/national governments in impacted nations

-Establish strong, reliable online presence-website, social media

-Taenacity is the only product of its kind, no vying for customers with competitors

-Get prominent medical journals and sites to review our device, expanding our audience and raising our reviews

-Use statistics + data about device efficacy

-Pair data with stories from local healthcare workers + real patients

Shreya

4. Cost Analysis

Manufacturing Costs Comparison

65K

Development Costs

Average development cost for a lateral flow assay

Source: S Rosen, NIH

$503

Current Cost for One Test

Cost of current test for cysticercosis caused by T. Solium

Source: NIH, R Bhattarai

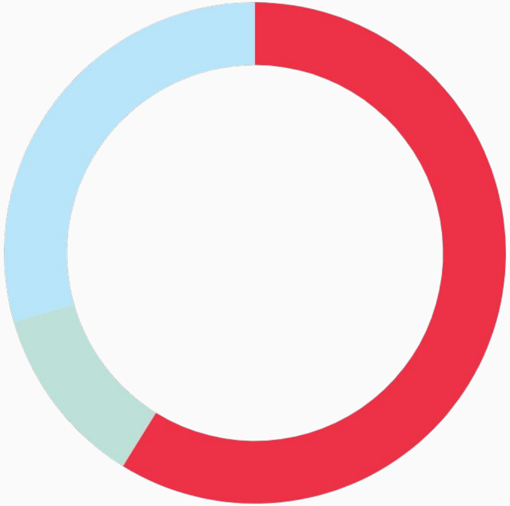
$3

Cost for our one Lateral flow Test

Cost when developed and manufactured

Source: Akram, Research Gate

Price Distribution Now



=

Profit: 42.3% (55K)

Companies invest in using our product to save money

Trials: 7.69% (10K)

Clinical trials

Product Development + Manufacturing: 50% (65K)

Buying materials

Salary Costs

CEO
$180,000

Business Analyst
$100,000

Marketing Specialist
$100,000

Regulatory Affairs Director
$120,000

Manufacturing Lead
$120,000

Head of R&D
$160,000

*based on CA avgs.

Market Growth/Sales Volume

$9.5 billion

Projected Market by 2029 from originally $5.9 billion

Source: Linkedin, Milestone Achieve Reports

8.17%

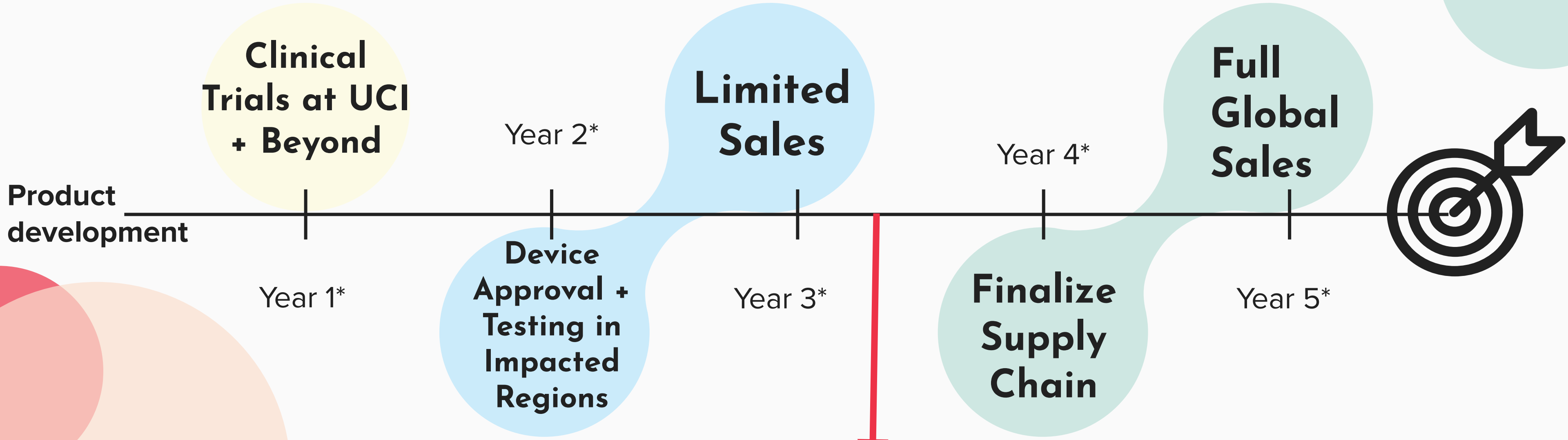
Annual Growth Rate of Market

Source: Linkedin, Milestone Achieve Reports

$17

Profit from each $20 test after manufacturing for 3$

Profit + Testing Timeline



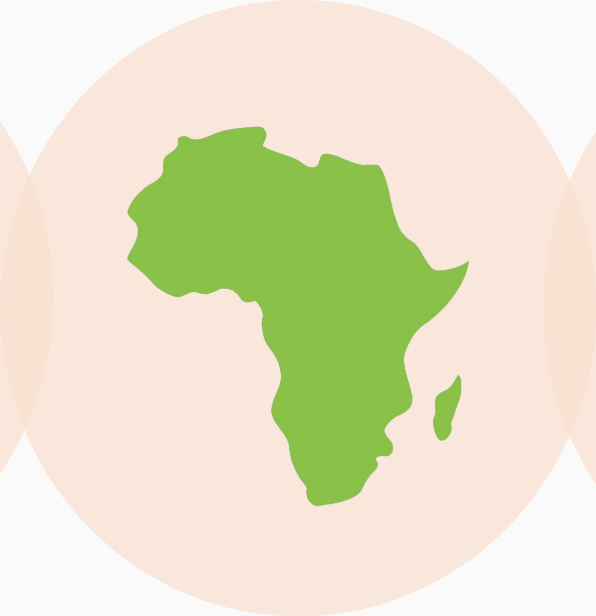
Profitability Achieved: ***76,000 unit sales***

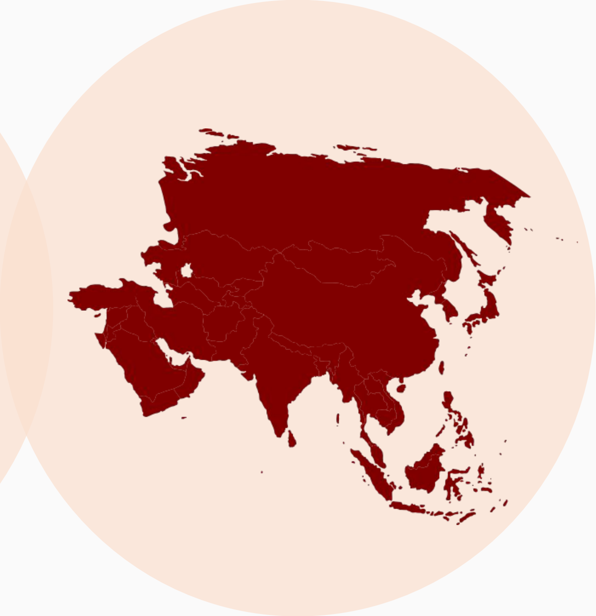
*Estimated Time

5. Summary

Prevalence

- Over **2.5 million people** are currently infected with T. Solium, with impoverished communities at risk in:





Our Device:

- A multiplexed lateral flow testing strip equipped with 150 nm gold nanoshells to increase device sensitivity and improve accuracy

Lab Processing

Instant Results

Cost-friendly

Our Need

$125,000 seed investment

- Manufacturing, Development, Testing
- Access to UCI Lab facilities/Resources

*Further ROI for our investors during our eventual IPO/public offering

Anticipated Annual Profit by Year 5: $31,500,000

*Achieved when mass production is reached w/ min. 2 million units per year→profitability by Year 3-3.5

Investors Receive: **15% equity** in LFAssociates + **15% share of profit** →**$472,500 annual profit**

Taenacity...

Fighting parasitic diseases **one test at a time...**

imart

Thanks

Do you have any questions?

LFAssociates